



symbolic-dag
closed form

$$R_1 \equiv I(X_1; Y \mid X_2), \quad R_2 \equiv I(X_2; Y \mid X_1) :$$

$$R_1 = \log \det(\Sigma_N + H_1 F_1 \Sigma_1 F_1^H H_1^H) - \log \det \Sigma_N$$

$$\frac{\partial R_1}{\partial F_1^*} = H_1^H (\Sigma_N + H_1 F_1 \Sigma_1 F_1^H H_1^H)^{-1} H_1 F_1 \Sigma_1$$

$$\frac{\partial R_2}{\partial F_2^*} = H_2^H (\Sigma_N + H_2 F_2 \Sigma_2 F_2^H H_2^H)^{-1} H_2 F_2 \Sigma_2$$